

Analyzing the evolution of scientific research topics on creatine supplementation.

Nicolas Forcella

Radboud University, Nijmegen, NL

ABSTRACT

Creatine supplementation has been studied since 1985 due to its health benefits, ranging from sports performance to clinical and neurological applications. As the volume of research grows, synthesizing and tracking thematic developments through manual reviews becomes increasingly challenging. This study applies topic modeling techniques to analyze the temporal evolution of research topics related to creatine supplementation. Using titles and abstracts retrieved from PubMed, a corpus of 806 research articles published between 1985 and 2025 were analyzed. Latent Dirichlet Allocation (LDA) was employed to identify research themes. Nineteen topics were identified and subsequently grouped into seven thematic clusters. By examining topic relative relevancy across five-year time bins, this study reveals a shift in research focus over time, from early emphasis on exercise performance and renal safety, toward emerging interests in neurological applications, musculoskeletal health, and aging. These findings demonstrate the utility of topic modeling for longitudinal analysis of creatine supplementation research trends and provide an overview of how scientific attention has evolved over time.

1 INTRODUCTION

With more than 800 research articles, creatine supplementation is a widely studied and utilized nutritional supplement. Initially popularized for its performance-enhancing effects on high-intensity exercise, creatine has since been investigated for a broad range of physiological and therapeutic applications, including neurological benefits, musculoskeletal health and aging. Over the past three decades, research on creatine supplementation has expanded substantially, producing a diverse body of literature.

Traditionally, the synthesis of scientific trends in this domain has relied on narrative reviews and expert-driven meta-analyses. While these approaches provide valuable insights, they are inherently labor-intensive and difficult to scale as publication volumes increase. Moreover, manual reviews require frequent updates to remain current and may overlook emerging or niche research themes. As a result, there is a growing need for automated, data-driven methods that can complement traditional reviews by systematically identifying and tracking research topics over time [18].

Topic modeling offers a computational solution to this challenge by uncovering semantic structures within large document collections. Latent Dirichlet Allocation (LDA), in particular, has been widely applied to analyze thematic trends in scientific literature across various domains, including medicine and public health [8, 18]. By representing documents as a distribution of topics, and topics as

distributions over words, LDA enables the exploration of research themes without requiring predefined categories.

In this project, topic modeling is applied to abstracts and titles of PubMed research articles related to creatine supplementation. The primary objective is to characterize how scientific focus in this field has evolved over time and to identify major thematic shifts.

2 RELATED WORK

2.1 Creatine Supplementation Topics

Creatine supplementation is widely used among athletes as it improves muscle mass, performance and recovery [3]. Other benefits of creatine include injury prevention, rare creatine-metabolizing syndromes, cognition and brain health, sarcopenia and bone health, heart failure and skin aging [10].

Moreover, two recent expert reviews synthesized research on creatine by examining a broad set of questions, including: (1) whether creatine leads to water retention; (2) whether it causes kidney damage or renal dysfunction; (3) whether it causes hair loss; (4) whether it leads to dehydration and muscle cramping (5) whether is it harmful for children and adolescents; (6) whether it increases fat mass; (7) whether it is beneficial for older adults; (8) whether it is only useful for resistance or power type activities; (9) whether it is only effective for males; (10) whether Creatine Monohydrate (CrM) and caffeine oppose each other; (11) whether CrM is anti-inflammatory; (12) whether CrM causes cancer; (13) whether CrM increases urine production; (14) whether CrM influences blood pressure; (15) whether CrM is safe to consume during pregnancy; (16) whether CrM enhances performance in adolescents; (17) whether CrM adversely affects male fertility; (18) whether CrM attenuates symptoms of sleep deprivation; and (19) whether CrM reduces the severity of and/or improve recovery from traumatic brain injury [2, 3].

These topics discussed provide a useful basis for further labeling and evaluation of the topics obtained by computational methods.

2.2 Topic Modeling

The standard approach towards analyzing research trends consists on manual reviews of the available research by experts. However, this approach is labor intensive and needs periodic updates to discover new research trends. One solution to this problem is topic modeling using computational techniques [18].

One widely used algorithm for achieving this is Latent Dirichlet Allocation (LDA). LDA is a probabilistic model for collections of discrete data. Each collection (in this case documents) is modeled as a probability distribution over topics; which are themselves modeled as a probability distribution of the words on the vocabulary of the corpus [5].

Since the posterior distribution of the LDA model given the observed words is intractable, different approximation methods must

be used, which can in turn yield different results [7]. In this paper we will focus in the two implementations found in the reviewed literature. The Gensim [14] python package implementation uses the variational Bayes approach which provides a faster but potentially biased implementation, in contrast with the MALLET [11] Java-based package, which uses Gibbs sampling potentially yielding more accurate results but requiring more computational resources [18].

Previous work have used this technique to model topics over time in the medical research domain. For instance, [8] used the MALLET implementation of LDA to assess research trends in the journal *Gynecologic Oncology*. In addition, [18] explored the use of the Gensim and the MALLET implementation of LDA to model emergency medicine research topics over time, obtaining better results with the MALLET implementation.

In this project, we applied LDA topic modeling to abstracts and titles related to creatine supplementation published in PubMed. We first explored the use of the Gensim's implementation and then shifted to MALLET as we struggled with convergence issues.

3 APPROACH

3.1 Study design

The methodological pipeline consisted of the following steps: (1) retrieval of relevant publications from PubMed, (2) text pre-processing, (3) topic model training, (4) topic visualization and evaluation, (5) topic labeling, and (6) analysis of topic trends over time.

Topic modeling outcomes are sensitive to data pre-processing choices, model implementations and hyperparameter selection. Consequently, the final configuration was selected through an iterative process aimed at maximizing topic coherence and interpretability. The sections below describe the explored alternatives and report the adopted methodologies and parameters for the final analysis.

3.2 Dataset retrieval

All scientific publications containing the term "creatine supplementation" and published on PubMed before the 15 of December were retrieved using biopython's Entrez module [6], yielding a total of 1089 publications. For each publication the title, abstract, journal, article type and journal year of publication were extracted.

Publications lacking abstract, title or year of publication were dropped. Each publication contains one or more associated types. If one of those types was 'Comment' or 'Published Erratum' the publication was removed. Articles were classified between research or review based on the article type following the criteria presented in Table 1, review articles were removed, resulting in 809 documents. Finally, the title and abstract text was concatenated to produce the corpus to be analyzed.

3.3 Data pre-processing

The following pre-processing techniques were explored as part of the iterative process for selecting the final pipeline:

Stopword removal: Stopwords are common, low-meaning words that do not contribute to the topics we intend to explore. The standard nltk [4] python package was used for tokenization and stopword removal. In addition, words such as "creatine", "supplementation" and "however" were also removed to prevent trivial topic dominance.

Lemmatization: Lemmatization is the process of reducing the words to their base word or lemma. This step transforms words into the same form, reducing the sparsity of the bag of words representation. The pre-processing pipeline leveraged the spaCy [9] "en_core_web_sm" model for lemmatization while keeping only the nouns, adjectives, adverbs and verbs.

Simple pre-processing: The texts were then lowercased, tokenized and de-accented using Gensim's [14] simple pre-processing method.

Bigrams and Trigrams: Bigrams are built by combining words that frequently appear together in the corpus into one token. Bigrams and Trigrams may allow to incorporate more meaningful topic words, while making the representation sparser. Bigrams and Trigrams were built using Gensim's [14] Phraser module with a threshold of 15. However, these representations reduced topic coherence and did not substantially improve interpretability; therefore, they were excluded from the final pipeline.

Common words removal: Tokens with a TF-IDF score below 0.01 were removed to further reduce noise.

3.4 Model Selection and Training

Initial experiments employed Gensim [14] LDA implementation. Models with topic counts ranging from 7 to 19 were trained and evaluated using the C_V coherence metric [15], averaged over ten runs to account for variability due to random initialization.

Across this range, coherence scores varied between approximately 0.43 and 0.45. High-scoring models were further examined qualitatively using pyLDavis [16] to assess topic separability and interpretability. Despite iterative tuning of pre-processing steps and hyperparameters, the Gensim-based models exhibited instability and limited consistency in smaller topics across runs.

Model training and evaluation were performed iteratively while experimenting with multiple pre-processing configurations and hyper-parameters. Despite these efforts, none of the explored configurations produced stable or semantically consistent topics across multiple runs, suggesting sensitivity to initialization and suboptimal convergence behavior.

Given prior evidence that Gibbs-sampling-based approaches can yield more stable topics [8, 18], subsequent experiments adopted the Java-based MALLET implementation [11] of LDA via the little_mallet_wrapper [1]. This approach produced higher coherence scores (ranging from 0.45 to 0.49) and more reproducible topic structures. Nineteen topics were selected for the final model, as this configuration maximized the amount of topics while maintaining coherence and interpretability.

3.5 Topic visualization and evaluation

Model performance was evaluated using the C_V topic coherence metric [15], as implemented in Gensim, which has been shown to

correlate strongly with human judgments of topic quality. Topic distributions and inter-topic relationships for Gensim's LDA models were visualized using pyLDAvis [16] to support qualitative analysis. Additionally, the top 20 words per topic were manually inspected. For the MALLET implementation, the five most representative documents per topic were also examined to assess semantic coherence.

3.6 Topic Labeling

Topic labeling was performed through manual inspection of the top 20 terms associated with each topic, informed by prior review literature on creatine supplementation. To assist this process, ChatGPT [13] was used to generate candidate labels based on topic keywords. These suggestions were reviewed and refined as necessary. After labeling, related topics were grouped into higher-level thematic clusters to facilitate trend analysis.

3.7 Topic trend calculation

Publications were grouped into five-year time bins, with the exception of the first bin, which spanned seven years due to the limited number of early publications. For each bin, topic proportions were aggregated and normalized by the number of documents to obtain a relative measure of topic prevalence. Topic proportions were then summed within thematic clusters to analyze higher-level research trends over time.

4 RESULTS AND ANALYSIS

4.1 Data description

After filtering, the final dataset consisted of 806 research articles. Following pre-processing, documents contained an average of 108.37 tokens. The earliest publication in the dataset appeared in 1985, after which research activity remained limited until the early 1990s. From approximately 1993 onward, the number of publications started to increase (Figure 2).

4.2 Model selection

The best-performing model employed the MALLET implementation of LDA with nineteen topics and default hyper-parameters, achieving a C_V coherence score of 0.47. The top 20 most probable words for each topic are presented in Table 2.

4.3 Topic labeling and clustering

The nineteen topics were labeled based on semantic inspection and prior literature and subsequently grouped into seven higher-level clusters: (1) Clinical trials and supplementation terms, (2) Exercise performance and sports science, (3) Bioenergetics and metabolic mechanisms, (4) Neurological and neurodevelopmental research, (5) Animal and cellular disease models, (6) Renal effects and creatinine metabolism, and (7) Musculoskeletal health and aging.

An overview of topic labels and cluster assignments is provided in Table 2.

4.4 Trend analysis

For each time bin, the relative prevalence of topics was computed and visualized (Figure 3). Topic proportions were then aggregated at the cluster level to examine broader research trends (Figure 1).

These results form the basis for the longitudinal analysis presented in the Discussion.

4.5 Limitations and Failure Analysis

Topic modeling is an unsupervised methodology that relies on qualitative interpretation, which inherently limits the ability to draw causal conclusions. In addition, both the MALLET and Gensim implementations of LDA rely on sampling-based inference methods that approximate the posterior distribution. As a result, model outputs exhibit variability across runs due to random initialization and sampling effects.

While most topics and high-probability terms remained relatively stable across multiple runs, some variability was observed, likely influenced by the limited size of the dataset. The results presented in this study correspond to a single representative run, consequently, topic proportions and trend analysis may vary across executions of the model. Furthermore, topic labeling and clustering are inherently subjective processes. In this study, labeling was performed by a single annotator with limited domain expertise, supported by a Large Language Model. This introduces potential bias in topic interpretation and cluster assignment.

Additionally, smaller or highly specific research themes may not be adequately captured due to data sparsity or may be absorbed into broader topics, limiting the resolution of niche research areas.

A specific issue observed across multiple runs was instability in topic proportions within the 1985–1991 time bin. This bin contains only a single document, making topic proportion estimates highly sensitive to stochastic variation. To examine this effect, a limited consistency analysis was conducted for the topics “renal function and creatinine excretion” and “creatine deficiency syndromes and neurological disorders” by re-running the model using identical parameters. While the overall trend shapes remained consistent, the extreme outlier observed in the first time bin disappeared. This suggests that the initial deviation is more likely an artifact of stochastic sampling rather than a meaningful signal. Consequently, this data point was excluded from trend interpretation and did not inform the study's conclusions.

5 DISCUSSION

The identified topics broadly align with themes reported in prior creatine research, supporting the validity of the modeling approach. However, larger and more established research areas were more clearly delineated, whereas smaller or niche topics tended to merge with broader themes. For example, topics related to urine production and blood pressure were incorporated into a broader renal function cluster rather than forming distinct topics. Similarly, research on caffeine-creatine interactions appeared only in eight documents and was not consistently captured across runs as a separate theme. Accordingly, the discussion focuses on the largest and most stable thematic clusters, supported by external literature that provides independent validation and plausible explanations for the observed temporal patterns.

Scientific interest in creatine supplementation began to expand markedly in the early 1990s, following studies demonstrating that oral creatine intake increases intramuscular creatine content and enhances high-intensity exercise performance, which contributed

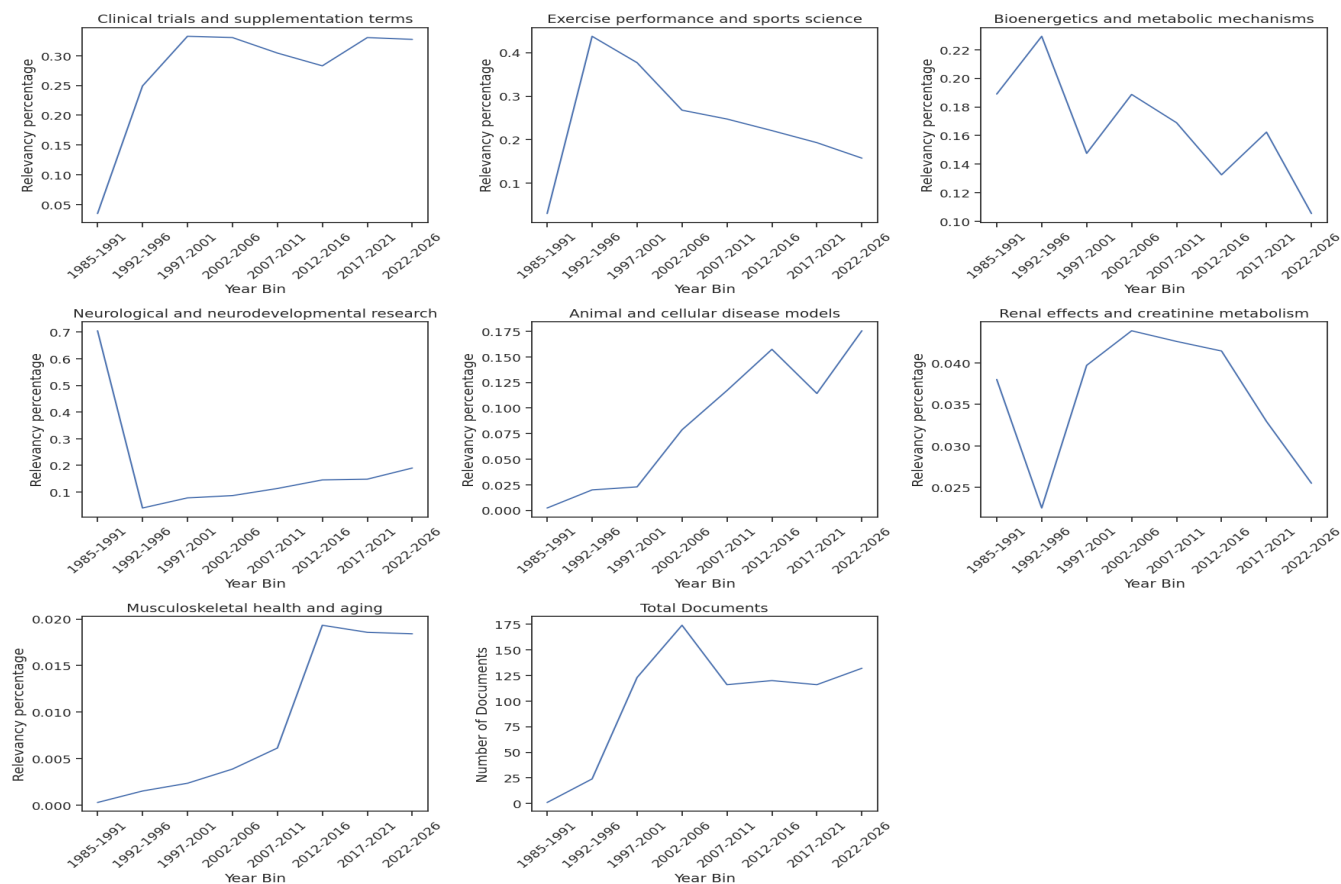


Figure 1: Cluster proportion over year bins.

to its widespread adoption in sport science [17]. This period coincides with the observed increase in publication volume beginning around 1993 (Figure 2).

Early in this expansion, concerns emerged regarding potential renal effects associated with creatine supplementation. These concerns prompted focused investigation, but were subsequently mitigated by accumulating evidence and meta-analyses reporting no significant adverse impact on kidney function in healthy populations [12]. Consistent with this narrative, the renal effects cluster exhibits increased research attention between approximately 1992 and 2016, followed by a decline. This pattern suggests a phase of heightened scrutiny that ultimately gave way to scientific consensus regarding renal safety.

A similar temporal pattern is observed for exercise and sport performance research, which peaks during the 1992–1996 period and then gradually declines. This trend may reflect the establishment of consensus regarding the performance-enhancing effects of creatine supplementation across various athletic disciplines, reducing the need for continued studies. As the main effect foundational questions were resolved, research attention appears to have shifted away from performance-focused outcomes.

In parallel, our results suggest that interest has grown in other domains, including potential neurological and neurodevelopmental

applications, creatine’s role in musculoskeletal health and potential benefits for aging population.

6 CONCLUSION AND OUTLOOK

This study analyzed the evolution of research topics on creatine supplementation using topic modeling techniques. The results indicate an initial phase characterized by rapid growth in research activity and a strong focus on exercise performance, accompanied by investigations into potential renal safety concerns. As consensus emerged regarding both the performance-related benefits and renal safety of creatine supplementation, research attention progressively shifted toward a more diverse set of topics, including neurological applications, musculoskeletal health, and aging-related outcomes.

Future work could strengthen these findings through the involvement of domain experts in topic labeling and evaluation, which would help reduce subjectivity and improve interpretability. Additional directions include exploring alternative topic coherence metrics, incorporating background corpora to improve topic stability and separation, and evaluating a broader range of topic counts or alternative topic modeling algorithms. Finally, the use of complementary visualization and clustering techniques may further enhance the analysis of longitudinal research trends.

REFERENCES

- [1] Maria Antoniak. 2024. little-mallet-wrapper. <https://github.com/maria-antoniak/little-mallet-wrapper>. Accessed: 2025-12-24.
- [2] J. Antonio, A. F. Brown, D. G. Candow, P. D. Chilibeck, S. J. Ellery, S. C. Forbes, B. Gualano, A. R. Jagim, C. Kerksick, R. B. Kreider, S. M. Ostojic, E. S. Rawson, M. D. Roberts, H. Roschel, A. E. Smith-Ryan, J. R. Stout, M. A. Tarnopolsky, T. A. VanDusseldorp, D. S. Willoughby, and T. N. Ziegenfuss. 2025. Part II. Common questions and misconceptions about creatine supplementation: what does the scientific evidence really show? *Journal of the International Society of Sports Nutrition* 22, 1 (2025), 2441760. doi:10.1080/15502783.2024.2441760
- [3] J. Antonio, D. G. Candow, S. C. Forbes, B. Gualano, A. R. Jagim, R. B. Kreider, E. S. Rawson, A. E. Smith-Ryan, T. A. VanDusseldorp, D. S. Willoughby, and T. N. Ziegenfuss. 2021. Common questions and misconceptions about creatine supplementation: what does the scientific evidence really show? *Journal of the International Society of Sports Nutrition* 18, 1 (2021), 13. doi:10.1186/s12970-021-00412-w
- [4] S. Bird and E. Loper. 2004. NLTK: The Natural Language Toolkit. In *Proceedings of the ACL Interactive Poster and Demonstration Sessions*. Association for Computational Linguistics, Barcelona, Spain, 214–217. <https://aclanthology.org/P04-3031/>
- [5] D. M. Blei, A. Y. Ng, and M. I. Jordan. 2003. Latent dirichlet allocation. *The Journal of Machine Learning Research* 3 (March 2003), 993–1022.
- [6] P. J. A. Cock, T. Antao, J. T. Chang, B. A. Chapman, C. J. Cox, A. Dalke, I. Friedberg, T. Hamelryck, F. Kauff, B. Wilczynski, and M. J. L. de Hoon. 2009. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics* 25, 11 (03 2009), 1422–1423. doi:10.1093/bioinformatics/btp163
- [7] W. M. Darling. 2011. A Theoretical and Practical Implementation Tutorial on Topic Modeling and Gibbs Sampling. In *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*. 642–647.
- [8] A. E. Grubbs, N. Sinha, R. Garg, and E. L. Barber. 2023. Use of topic modeling to assess research trends in the journal *Gynecologic Oncology*. *Gynecologic Oncology* 172 (2023), 41–46. doi:10.1016/j.ygyno.2023.03.001
- [9] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd. 2020. spaCy: Industrial-strength Natural Language Processing in Python. *Zenodo* (2020). doi:10.5281/zenodo.1212303
- [10] Mayo Clinic. 2023. *Creatine*. Retrieved December 12, 2025 from <https://www.mayoclinic.org/drugs-supplements-creatine/art-20347591>
- [11] A. K. McCallum. 2002. *MALLET: A machine learning for Language Toolkit*. Retrieved December 25, 2025 from <http://mallet.cs.umass.edu>
- [12] Ehsan Khorasani Naeini, Mahsa Eskandari, Mohsen Mortazavi, Ali Gholaminejad, and Nima Karevan. 2025. Effect of creatine supplementation on kidney function: a systematic review and meta-analysis. *BMC Nephrology* 26, 1 (2025), 622. doi:10.1186/s12882-025-04558-6
- [13] OpenAI. 2025. ChatGPT (Dec 25 version). Large Language Model.
- [14] R. Řehůřek and P. Sojka. 2010. Software Framework for Topic Modelling with Large Corpora. In *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*. ELRA, Valletta, Malta, 45–50.
- [15] M. Röder, A. S. Both, and A. Hinneburg. 2015. Exploring the Space of Topic Coherence Measures. In *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining* (Shanghai, China) (WSDM '15). Association for Computing Machinery, New York, NY, USA, 399–408. doi:10.1145/2684822.2685324
- [16] C. Sievert and K. Shirley. 2014. LDAvis: A method for visualizing and interpreting topics. In *Proceedings of the Workshop on Interactive Language Learning, Visualization, and Interfaces*. Association for Computational Linguistics, 63–70. doi:10.3115/v1/W14-3110
- [17] J. R. Stout, R. B. Kreider, D. G. Candow, S. C. Forbes, E. S. Rawson, B. Antonio, and J. Antonio. 2025. The birth of modern sports nutrition: tracing the path from muscle biopsies to creatine supplementation—A narrative review. *Journal of the International Society of Sports Nutrition* 22, sup1 (2025), 2463373. arXiv:<https://doi.org/10.1080/15502783.2025.2463373> doi:10.1080/15502783.2025.2463373 PMID: 39964101.
- [18] P. Thomas and R. A. Taylor. 2021. Forty years of emergency medicine research: Uncovering research themes and trends through topic modeling. *The American Journal of Emergency Medicine* 45 (2021), 213–220. doi:10.1016/j.ajem.2020.08.036

7 APPENDIX

7.1 Work Report

We began by reviewing related research articles that applied topic modeling in the medical domain. We then examined review articles on creatine supplementation that described existing research topics and trends, which served as a basis for validation. Next, we explored Python packages for topic modeling, text pre-processing, and data retrieval from PubMed.

I subsequently refined and evaluated search terms in the PubMed web search to ensure that the resulting corpus was restricted to publications specifically focused on creatine supplementation. We then implemented the pre-processing pipelines and topic modeling using Gensim, iterating through different configurations until no satisfactory convergence was achieved.

Given these limitations, we explored the MALLET implementation. Because the Gensim wrapper for MALLET had been deprecated, we identified an alternative wrapper and reviewed its source code to ensure suitability. As MALLET produced more coherent and semantically related topic word groupings, we reimplemented the code originally developed for the Gensim-based approach, with the exception of pyLDavis, whose integration required substantial additional effort.

The report was written while collecting the information and once stable results were achieved. Through the writing process Generative AI was used to assist with writing style, flow, and word choice.

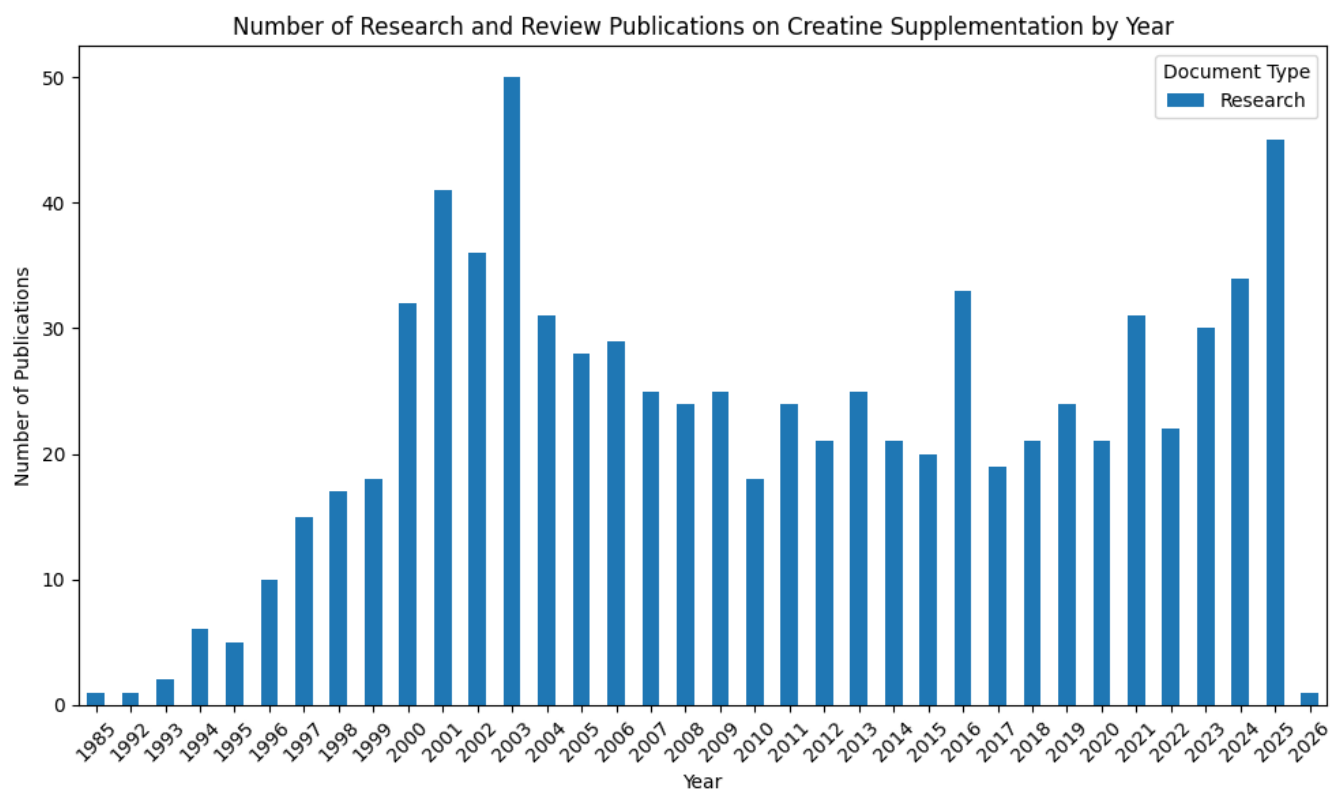


Figure 2: Number of publications on creatine supplementation per year.

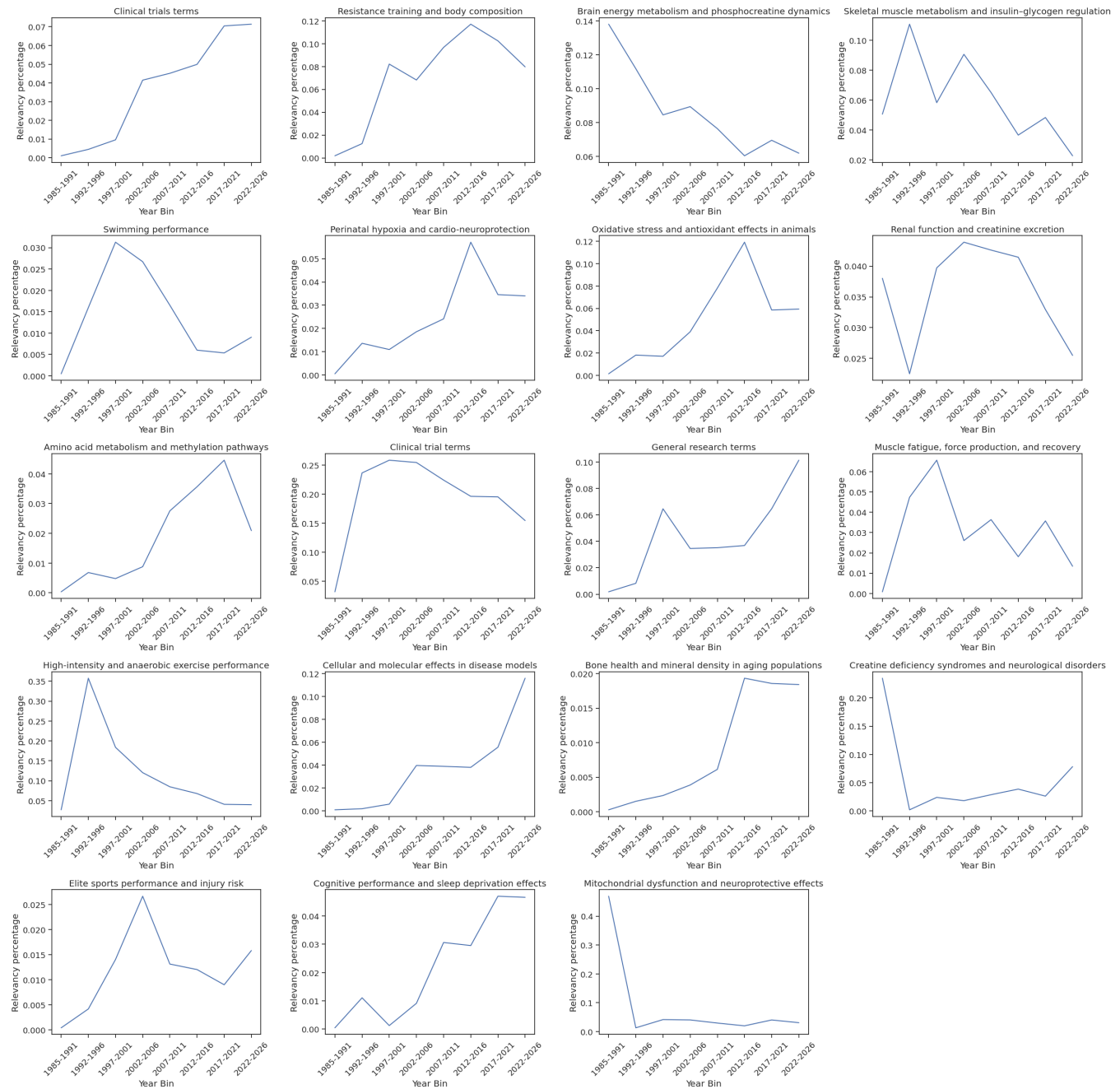


Figure 3: Topic proportion over year bins.

Article type	Classification
Case Reports	Research
Clinical Study	Research
Clinical Trial	Research
Clinical Trial Protocol	Research
Clinical Trial, Veterinary	Research
Comparative Study	Research
Controlled Clinical Trial	Research
Editorial	Review
English Abstract	Review
Evaluation Study	Research
Historical Article	Review
Journal Article	Research
Meta-Analysis	Review
Multicenter Study	Research
Preprint	Research
Randomized Controlled Trial	Research
Research Support, N.I.H., Extramural	Research
Research Support, Non-U.S. Gov't	Research
Research Support, U.S. Gov't, Non-P.H.S.	Research
Research Support, U.S. Gov't, P.H.S.	Research
Review	Review
Scoping Review	Review
Systematic Review	Review
Validation Study	Research

Table 1: Article publication types and their classification between Research and Reviews

Topic Id	Topic words	Topic label	Cluster
1	patient muscle placebo function trial week strength control effect blind improve study double outcome functional randomize assess quality physical life	Clinical trials terms	Clinical trials and supplementation terms
2	training resistance strength body group mass muscle week increase repetition press composition fat effect leg exercise placebo train performance endurance	Resistance training and body composition	Exercise performance and sports science
3	increase energy concentration level brain phosphocreatine suggest high activity human result phosphate decrease metabolism oral uptake ratio show use lead	Brain energy metabolism and phosphocreatine dynamics	Bioenergetics and metabolic mechanisms
4	muscle content increase skeletal fiber glucose protein diet expression glycogen total effect beta feed insulin study type control rat dietary	Skeletal muscle metabolism and insulin-glycogen regulation	Bioenergetics and metabolic mechanisms
5	rest minute exercise swimmer swimming pressure postexercise compartment temperature swim heat male body bout leg response hormone blood anterior complex	Swimming performance	Exercise performance and sports science
6	mouse heart hypoxia fetal birth cardiac asphyxia injury function reduce maternal ischemia brain treatment pregnancy cerebral acute term cord protect	Perinatal hypoxia and cardio-neuroprotection	Neurological and neurodevelopmental research
7	rat effect group stress oxidative induce exercise diet activity antioxidant animal liver study supplement decrease marker lipid week level damage	Oxidative stress and antioxidant effects in animals	Animal and cellular disease models
8	creatinine kidney renal blood function plasma urine concentration urinary serum supplement rate level sample excretion diet oral athlete term urea	Renal function and creatinine excretion	Renal effects and creatinine metabolism
9	plasma concentration math homocysteine level arginine acid low decrease increase synthesis group reduce methylation effect lung crs participant metabolite receive	Amino acid metabolism and methylation pathways	Bioenergetics and metabolic mechanisms
10	group day effect study increase significant subject follow placebo body time monohydrate difference change post use compare result control measure	Clinical trial terms	Clinical trials and supplementation terms
11	use supplement high study health report potential effect athlete analysis dietary evidence need method performance base side sport adverse nutritional	General research terms	Clinical trials and supplementation terms
12	muscle fatigue exercise contraction force torque recovery knee strength isometric voluntary isokinetic peak maximal damage eccentric induce skeletal measure extensor	Muscle fatigue, force production, and recovery	Exercise performance and sports science
13	exercise performance sprint power test maximal perform trial placebo effect repeat intensity improve high peak bout anaerobic lactate min work	High-intensity and anaerobic exercise performance	Exercise performance and sports science
14	cell mouse tumor induce expression increase role disease result inflammatory protein tissue neuron enhance study effect model system significantly culture	Cellular and molecular effects in disease models	Animal and cellular disease models
15	bone mineral old density elderly content month tissue health parameter postmenopausal spine year age ovariectomize femoral level proteomic femur property	Bone health and mineral density in aging populations	Musculoskeletal health and aging
16	deficiency patient seizure treatment glycine gamt year arginine brain disorder show epilepsy mutation syndrome transporter clinical diagnosis delay low cerebral	Creatine deficiency syndromes and neurological disorders	Neurological and neurodevelopmental research
17	player athlete football performance injury soccer season elite muscle specific condition highly subgroup jump caffeine incidence skill jumping adolescent low	Elite sports performance and injury risk	Exercise performance and sports science
18	cognitive brain effect sleep memory test female performance deprivation male adult improve chronic week administration like depression participant behavior deficit	Cognitive performance and sleep deprivation effects	Neurological and neurodevelopmental research
19	patient treatment disease muscle mouse model mitochondrial effect improve motor treat significantly age dysfunction administration neuroprotective weight symptom skeletal increase	Mitochondrial dysfunction and neuroprotective effects	Neurological and neurodevelopmental research

Table 2: Topic modeling results showing top words, inferred semantic labels, and higher-level thematic clusters